

## Nonlinear Programming (NLP)

### General function

objective function  $\max f(x_1, x_2, x_3, \dots, x_n)$

$$s.t. \begin{cases} g_1(x_1, x_2, x_3, \dots, x_n) \leq b_1 \\ g_2(x_1, x_2, x_3, \dots, x_n) \leq b_2 \\ \dots \\ g_m(x_1, x_2, x_3, \dots, x_n) \leq b_m \end{cases}$$

$$g_i(x_1, x_2, x_3, \dots, x_n) = \sum_{j=1}^n a_{ij}x_j, \text{ where } i = 1, 2, 3, \dots, m$$

### Nonlinear Programming

objective function  $\max \left( \min \right) y = (\text{a nonlinear combination of independent variables})$

$s.t.$  all the constraints

notice that for some programming, the objective function can only be calculated as the maximum, hence need to [transform the minimum function into a maximum function](#), by adding a negative sign

e.g. An investor has \$5000 and two potential investments. Let  $x_j$  for  $j = 1$  and  $j = 2$

denote his allocation to investment  $j$  in thousands of dollars. From historical data, investments 1 and 2 have an expected annual return of 20 and 16 percent, respectively. Also, the total risk involved with investments 1 and 2, as measured by the variance of total return, is given by:  $2x_1^2 + x_2^2 + (x_1 + x_2)^2$ .

so that risk increases with total investment and with the amount of each individual investment. The investor would like to maximise his expected return and at the same time minimise his risk.

$$\max f(x) = 20x_1 + 16x_2 - \theta[2x_1^2 + x_2^2 + (x_1 + x_2)^2]$$

$$\begin{cases} g_1(x) = x_1 + x_2 \leq 50 \\ x_1 \geq 0, x_2 \geq 0 \text{ (that is, } g_2(x) = -x_1, g_3(x) = -x_2) \end{cases}$$

where the nonnegative constant  $\theta$  reflects his trade off between risk and return

